# Credit Card Approval Prediction

## Phase 1: Brainstorming & Ideation

SmartBridge / SkillWallet Virtual Internship Program

| Team Member | Role |
|---|---|
| Tupakula Rishmitha Sri | Team Lead |
| Tejomai Kondapureddy | Member |
| Korrapolu Lahari | Member |
| Saragada Tirupathi Venkata Mohan Reddy | Member |
| Vijay Kanth Kalvakuntla | Member |

## 1. Problem Statement

Banks and financial institutions receive thousands of credit card applications every day. A significant portion are rejected due to factors such as high existing loan balances, insufficient income levels, or excessive credit inquiries. Manually reviewing each application is time-consuming and prone to human error, making it inefficient at scale. There is a need for an automated, data-driven system that can instantly assess applicant eligibility using historical financial and demographic patterns.

## 2. Brainstorming Session Summary

The team conducted a structured brainstorming session to explore possible approaches to automating credit card approval decisions. Ideas were grouped by theme and evaluated for feasibility, impact, and alignment with available data.

### 2.1 Ideas Generated

- Build a rule-based approval engine using fixed income/credit thresholds.
- Train a machine learning classifier on historical applicant and credit history data.
- Use a scoring system (credit score style) instead of a binary decision.
- Combine ML prediction with manual override for borderline cases.
- Provide a self-service eligibility checker for prospective customers.
- Build a compliance dashboard for batch-screening high-risk applicants.

### 2.2 Idea Prioritization

The team selected the machine learning classification approach as the primary solution, since it can learn complex, non-linear patterns from real applicant data that fixed rules would miss, while still supporting a manual-override workflow for edge cases. The self-service eligibility checker and compliance batch-screening ideas were retained as secondary use cases for the same underlying model.

## 3. Proposed Solution

An AI-powered Credit Card Approval Prediction System that automates eligibility decisions using machine learning. The model is trained on historical applicant information — income, employment status, education, marital status, and credit history — to predict whether an application should be Approved or Rejected. The solution is delivered through a Flask web application, with an optional IBM Watson Machine Learning deployment path for scalable cloud inference.

## 4. Key Features Identified

- Automated Approved / Rejected prediction with a confidence score.
- Comparison of multiple ML algorithms to select the best performer.
- Real-time prediction through a simple, responsive web form.
- Feature engineering from raw credit history into interpretable risk indicators (EMI Paid Off, EMI Pastdues, Number of Loans).
- Model serialization (Pickle) so predictions don't require retraining.
- Optional cloud deployment via IBM Watson Machine Learning.

## 5. Target Users

- **Credit Analysts** — get instant decisions to prioritize manual review.
- **Compliance Officers** — batch-screen high-risk applicants.
- **Prospective Customers** — self-check eligibility before formally applying.